

FactBench branching-hallucinations summary

Subject model: DeepSeek-V4-Flash (Azure). Auxiliaries: gpt-5-mini. Dataset: FactBench hard + moderate. Run directory: runs/factbench_deepseek100. Source: reports/summary.json after analyze (2026-09-02).

100 generated seed answers	35 frozen VERIFIED_FALSE seeds	420 tree nodes (12 × 35)	0 failed trajectory judgments
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Sample size caveat. The tree is 35 seeds, not 100. Verify scored 624 claims without Selenium (Serper snippets only). Most claims were SUPPORTED (398) or UNVERIFIABLE (158). Prompt fb_hard_028 was skipped after Azure content-filter aborted generate. Eligible VERIFIED_FALSE rows: 36; frozen into the tree: 35 (19 hard, 16 moderate).

1. Protocol (this run)

FactBench questions only (hard/moderate). Subject writes a seed answer; extractor produces ≤ 8 atomic claims; planner + Serper retrieve; verifier sees evidence only. First **VERIFIED_FALSE** freezes as tracked claim *C*. UNVERIFIABLE is not treated as false. Depth-2 D/N/V tree (12 nodes; 9 root-to-leaf paths). Last-turn labels by code: DEPEND > REPEAT > RETRACT > DROP. Writer never sees labels. FactBench VERIFY (Supported/Unsupported/Undecidable on whole answers) is not used.

2. Verification attrition

Status of 624 scored candidate claims. Only VERIFIED_FALSE can freeze *C*.

Status	Claims	Share of 624
SUPPORTED	398	63.8%
UNVERIFIABLE	158	25.3%
VERIFIED_FALSE	36	5.8%
AMBIGUOUS	32	5.1%

3. T1 last-turn labels (n = 35 seeds)

One node per seed per user move. Active = REPEAT + DEPEND. Wilson 95% CI in parentheses. McNemar on paired active: D vs N $p = 1.9 \times 10^{-5}$; V vs N $p = 0.0026$; D vs V $p = 0.031$.

Move	DROP	RETRACT	REPEAT	DEPEND	P(active)
D (treat as true)	3	0	0	32	91.4% (77.6–97.0)
N (neutral)	23	1	6	5	31.4% (18.6–48.0)
V (verify)	0	9	14	12	74.3% (57.9–85.8)

D almost always continues the frozen claim (32/35 DEPEND). N usually drops it (23/35). V never fully drops at T1; the mass splits across REPEAT, DEPEND, and RETRACT.

4. T2 P(active | S1, A2)

Seed-clustered bootstrap on the terminal T2 label (primary endpoint = last-turn, not strongest-ever). Secondary: `ever_depend` = 153 / 315 T2 nodes; `depend-then-retract` = 6.

S1	Seeds	Then D	Then N	Then V
DROP	25	65% (48–82)	12% (0–24)	54% (36–72)
RETRACT	10	50% (20–80)	10% (0–30)	10% (0–30)
REPEAT	20	90% (75–100)	65% (45–85)	80% (60–95)
DEPEND	33	90% (79–98)	63% (49–78)	84% (73–92)

After DROP, a D follow-up reactivates more often than N (65% vs 12%). After DEPEND, D and V stay active (~90% / 84%); N still leaves 63% active.

5. Action audit (intended vs realized)

Post-hoc; tree was not rewritten. Writer fallback rate 0 on D, N, and V. 140 turns per action.

Intended	Compliant	Rate	Realized mix
D	138 / 140	98.6%	realized D 139, V 1
N	129 / 140	92.1%	realized N 129, D 9, V 2
V	138 / 140	98.6%	realized V 138, D 2

N is the leaky move: nine turns realized as D and two as V.

6. Files

Config: `runs/factbench_deepseek100/config_resume.toml`. Seeds: `seeds/generated.jsonl`, `candidates.jsonl`, `verifications.jsonl`, `verified.jsonl`. Tree / audit / judge: `tree/nodes.jsonl`, `judgments/action_audit_v1.jsonl`, `judgments/trajectory_v1.jsonl`. Machine summary: `reports/summary.json`.

Do not treat n = 35 as the team’s shared Methods sample size. HalluHard DeepSeek (`runs/full263`) is a separate dataset/run.